

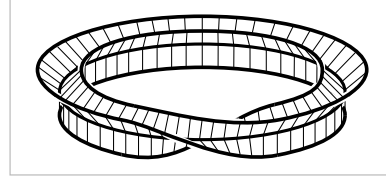
identified via the word a^2c^2 . By the proposition, $\pi_1(N_g) \approx \langle a_1, \dots, a_g \mid a_1^2 \cdots a_g^2 \rangle$. This abelianizes to the direct sum of $\mathbb{Z}_2$ with $g - 1$ copies of $\mathbb{Z}$ since in the abelianization we can rechoose the generators to be $a_1, \dots, a_{g-1}$ and $a_1 + \cdots + a_g$, with $2(a_1 + \cdots + a_g) = 0$. Hence N_g is not homotopy equivalent to N_h if $g \neq h$, nor is N_g homotopy equivalent to any orientable surface M_h .

Here is another application of the preceding proposition:

Corollary 1.28. *For every group G there is a 2-dimensional cell complex X_G with $\pi_1(X_G) \approx G$.*

Proof: Choose a presentation $G = \langle g_\alpha \mid r_\beta \rangle$. This exists since every group is a quotient of a free group, so the g_α 's can be taken to be the generators of this free group with the r_β 's generators of the kernel of the map from the free group to G . Now construct X_G from $\bigvee_\alpha S_\alpha^1$ by attaching 2-cells e_β^2 by the loops specified by the words r_β . $\square$

Example 1.29. If $G = \langle a \mid a^n \rangle = \mathbb{Z}_n$ then X_G is S^1 with a cell e^2 attached by the map $z \mapsto z^n$, thinking of S^1 as the unit circle in $\mathbb{C}$. When $n = 2$ we get $X_G = \mathbb{RP}^2$, but for $n > 2$ the space X_G is not a surface since there are n 'sheets' of e^2 attached at each point of the circle $S^1 \subset X_G$. For example, when $n = 3$ one can construct a neighborhood N of S^1 in X_G by taking the product of the graph Υ with the interval I , and then identifying the two ends of this product via a one-third twist as shown in the figure. The boundary of N consists of a single circle, formed by the three endpoints of each Υ cross section of N . To complete the construction of X_G from N one attaches a disk along the boundary circle of N . This cannot be done in $\mathbb{R}^3$, though it can in $\mathbb{R}^4$. For $n = 4$ one would use the graph $\times$ instead of Υ , with a one-quarter twist instead of a one-third twist. For larger n one would use an n -pointed 'asterisk' and a $1/n$ twist.

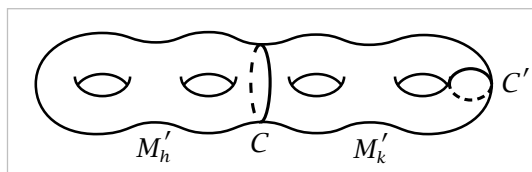


Exercises

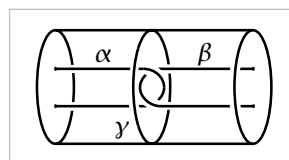
1. Show that the free product $G * H$ of nontrivial groups G and H has trivial center, and that the only elements of $G * H$ of finite order are the conjugates of finite-order elements of G and H .
2. Let $X \subset \mathbb{R}^m$ be the union of convex open sets $X_1, \dots, X_n$ such that $X_i \cap X_j \cap X_k \neq \emptyset$ for all i, j, k . Show that X is simply-connected.
3. Show that the complement of a finite set of points in $\mathbb{R}^n$ is simply-connected if $n \geq 3$.

4. Let $X \subset \mathbb{R}^3$ be the union of n lines through the origin. Compute $\pi_1(\mathbb{R}^3 - X)$.
5. Let $X \subset \mathbb{R}^2$ be a connected graph that is the union of a finite number of straight line segments. Show that $\pi_1(X)$ is free with a basis consisting of loops formed by the boundaries of the bounded complementary regions of X , joined to a basepoint by suitably chosen paths in X . [Assume the Jordan curve theorem for polygonal simple closed curves, which is equivalent to the case that X is homeomorphic to S^1 .]
6. Suppose a space Y is obtained from a path-connected subspace X by attaching n -cells for a fixed $n \geq 3$. Show that the inclusion $X \hookrightarrow Y$ induces an isomorphism on π_1 . [See the proof of Proposition 1.26.] Apply this to show that the complement of a discrete subspace of $\mathbb{R}^n$ is simply-connected if $n \geq 3$.
7. Let X be the quotient space of S^2 obtained by identifying the north and south poles to a single point. Put a cell complex structure on X and use this to compute $\pi_1(X)$.
8. Compute the fundamental group of the space obtained from two tori $S^1 \times S^1$ by identifying a circle $S^1 \times \{x_0\}$ in one torus with the corresponding circle $S^1 \times \{x_0\}$ in the other torus.

9. In the surface M_g of genus g , let C be a circle that separates M_g into two compact subsurfaces M'_h and M'_k obtained from the closed surfaces M_h and M_k by deleting an open disk from each. Show that M'_h does not retract onto its boundary circle C , and hence M_g does not retract onto C . [Hint: abelianize π_1 .] But show that M_g does retract onto the nonseparating circle C' in the figure.

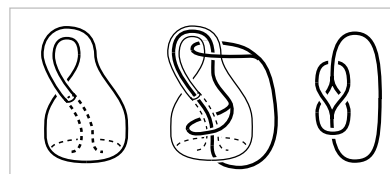


10. Consider two arcs α and β embedded in $D^2 \times I$ as shown in the figure. The loop γ is obviously nullhomotopic in $D^2 \times I$, but show that there is no nullhomotopy of γ in the complement of $\alpha \cup \beta$.



11. The **mapping torus** T_f of a map $f: X \rightarrow X$ is the quotient of $X \times I$ obtained by identifying each point $(x, 0)$ with $(f(x), 1)$. In the case $X = S^1 \vee S^1$ with f basepoint-preserving, compute a presentation for $\pi_1(T_f)$ in terms of the induced map $f_*: \pi_1(X) \rightarrow \pi_1(X)$. Do the same when $X = S^1 \times S^1$. [One way to do this is to regard T_f as built from $X \vee S^1$ by attaching cells.]

12. The Klein bottle is usually pictured as a subspace of $\mathbb{R}^3$ like the subspace $X \subset \mathbb{R}^3$ shown in the first figure at the right. If one wanted a model that could actually function as a bottle, one would delete the open disk bounded by the circle of self-intersection of X , producing a subspace $Y \subset X$. Show that $\pi_1(X) \approx \mathbb{Z} * \mathbb{Z}$ and that



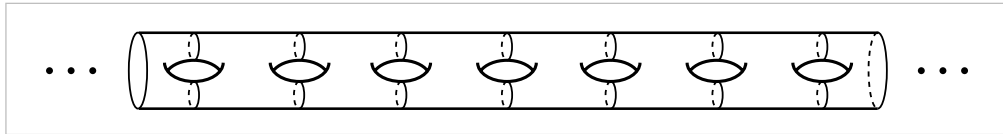
$\pi_1(Y)$ has the presentation $\langle a, b, c \mid aba^{-1}b^{-1}cb^\varepsilon c^{-1} \rangle$ for $\varepsilon = \pm 1$. (Changing the sign of ε gives an isomorphic group, as it happens.) Show also that $\pi_1(Y)$ is isomorphic to $\pi_1(\mathbb{R}^3 - Z)$ for Z the graph shown in the figure. The groups $\pi_1(X)$ and $\pi_1(Y)$ are not isomorphic, but this is not easy to prove; see the discussion in Example 1B.13.

13. The space Y in the preceding exercise can be obtained from a disk with two holes by identifying its three boundary circles. There are only two essentially different ways of identifying the three boundary circles. Show that the other way yields a space Z with $\pi_1(Z)$ not isomorphic to $\pi_1(Y)$. [Abelianize the fundamental groups to show they are not isomorphic.]

14. Consider the quotient space of a cube I^3 obtained by identifying each square face with the opposite square face via the right-handed screw motion consisting of a translation by one unit in the direction perpendicular to the face combined with a one-quarter twist of the face about its center point. Show this quotient space X is a cell complex with two 0-cells, four 1-cells, three 2-cells, and one 3-cell. Using this structure, show that $\pi_1(X)$ is the quaternion group $\{\pm 1, \pm i, \pm j, \pm k\}$, of order eight.

15. Given a space X with basepoint $x_0 \in X$, we may construct a CW complex $L(X)$ having a single 0-cell, a 1-cell e_γ^1 for each loop γ in X based at x_0 , and a 2-cell e_τ^2 for each map τ of a standard triangle PQR into X taking the three vertices P , Q , and R of the triangle to x_0 . The 2-cell e_τ^2 is attached to the three 1-cells that are the loops obtained by restricting τ to the three oriented edges PQ , PR , and QR . Show that the natural map $L(X) \rightarrow X$ induces an isomorphism $\pi_1(L(X)) \approx \pi_1(X, x_0)$.

16. Show that the fundamental group of the surface of infinite genus shown below is free on an infinite number of generators.



17. Show that $\pi_1(\mathbb{R}^2 - \mathbb{Q}^2)$ is uncountable.

18. In this problem we use the notions of suspension, reduced suspension, cone, and mapping cone defined in Chapter 0. Let X be the subspace of $\mathbb{R}$ consisting of the sequence $1, 1/2, 1/3, 1/4, \dots$ together with its limit point 0.

- (a) For the suspension SX , show that $\pi_1(SX)$ is free on a countably infinite set of generators, and deduce that $\pi_1(SX)$ is countable. In contrast to this, the reduced suspension ΣX , obtained from SX by collapsing the segment $\{0\} \times I$ to a point, is the shrinking wedge of circles in Example 1.25, with an uncountable fundamental group.
- (b) Let C be the mapping cone of the quotient map $SX \rightarrow \Sigma X$. Show that $\pi_1(C)$ is uncountable by constructing a homomorphism from $\pi_1(C)$ onto $\prod_{\infty} \mathbb{Z} / \bigoplus_{\infty} \mathbb{Z}$. Note

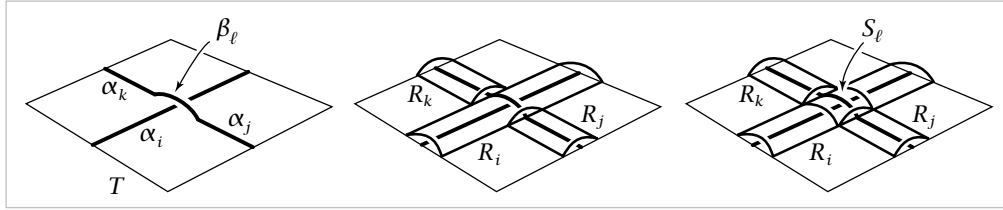
that C is the reduced suspension of the cone CX . Thus the reduced suspension of a contractible space need not be contractible, unlike the unreduced suspension.

19. Show that the subspace of $\mathbb{R}^3$ that is the union of the spheres of radius $1/n$ and center $(1/n, 0, 0)$ for $n = 1, 2, \dots$ is simply-connected.

20. Let X be the subspace of $\mathbb{R}^2$ that is the union of the circles C_n of radius n and center $(n, 0)$ for $n = 1, 2, \dots$. Show that $\pi_1(X)$ is the free group $*_n \pi_1(C_n)$, the same as for the infinite wedge sum $\bigvee_{\infty} S^1$. Show that X and $\bigvee_{\infty} S^1$ are in fact homotopy equivalent, but not homeomorphic.

21. Show that the join $X * Y$ of two nonempty spaces X and Y is simply-connected if X is path-connected.

22. In this exercise we describe an algorithm for computing a presentation of the fundamental group of the complement of a smooth or piecewise linear knot K in $\mathbb{R}^3$, called the *Wirtinger presentation*. To begin, we position the knot to lie almost flat on a table, so that K consists of finitely many disjoint arcs α_i where it intersects the table top together with finitely many disjoint arcs β_ℓ where K crosses over itself. The configuration at such a crossing is shown in the first figure below. We build a



2-dimensional complex X that is a deformation retract of $\mathbb{R}^3 - K$ by the following three steps. First, start with the rectangle T formed by the table top. Next, just above each arc α_i place a long, thin rectangular strip R_i , curved to run parallel to α_i along the full length of α_i and arched so that the two long edges of R_i are identified with points of T , as in the second figure. Any arcs β_ℓ that cross over α_i are positioned to lie in R_i . Finally, over each arc β_ℓ put a square S_ℓ , bent downward along its four edges so that these edges are identified with points of three strips R_i , R_j , and R_k as in the third figure; namely, two opposite edges of S_ℓ are identified with short edges of R_j and R_k and the other two opposite edges of S_ℓ are identified with two arcs crossing the interior of R_i . The knot K is now a subspace of X , but after we lift K up slightly into the complement of X , it becomes evident that X is a deformation retract of $\mathbb{R}^3 - K$.

- Assuming this bit of geometry, show that $\pi_1(\mathbb{R}^3 - K)$ has a presentation with one generator x_i for each strip R_i and one relation of the form $x_i x_j x_i^{-1} = x_k$ for each square S_ℓ , where the indices are as in the figures above. [To get the correct signs it is helpful to use an orientation of K .]
- Use this presentation to show that the abelianization of $\pi_1(\mathbb{R}^3 - K)$ is $\mathbb{Z}$.

1.3 Covering Spaces

We come now to the second main topic of this chapter, covering spaces. We have in fact already encountered one example of a covering space in our calculation of $\pi_1(S^1)$. This was the map $\mathbb{R} \rightarrow S^1$ that we pictured as the projection of a helix onto a circle, with the helix lying above the circle, ‘covering’ it. A number of things we proved for this covering space are valid for all covering spaces, and this allows covering spaces to serve as a useful general tool for calculating fundamental groups. But the connection between the fundamental group and covering spaces runs much deeper than this, and in many ways they can be regarded as two viewpoints toward the same thing. This means that algebraic features of the fundamental group can often be translated into the geometric language of covering spaces. This is exemplified in one of the main results in this section, giving an exact correspondence between the various connected covering spaces of a given space X and subgroups of $\pi_1(X)$. This is strikingly reminiscent of Galois theory, with its correspondence between field extensions and subgroups of the Galois group.

Let us begin with the definition. A **covering space** of a space X is a space $\tilde{X}$ together with a map $p: \tilde{X} \rightarrow X$ satisfying the following condition: There exists an open cover $\{U_\alpha\}$ of X such that for each α , $p^{-1}(U_\alpha)$ is a disjoint union of open sets in $\tilde{X}$, each of which is mapped by p homeomorphically onto U_α . We do not require $p^{-1}(U_\alpha)$ to be nonempty, so p need not be surjective.

In the helix example one has $p: \mathbb{R} \rightarrow S^1$ given by $p(t) = (\cos 2\pi t, \sin 2\pi t)$, and the cover $\{U_\alpha\}$ can be taken to consist of any two open arcs whose union is S^1 . A related example is the helicoid surface $S \subset \mathbb{R}^3$ consisting of points of the form $(s \cos 2\pi t, s \sin 2\pi t, t)$ for $(s, t) \in (0, \infty) \times \mathbb{R}$. This projects onto $\mathbb{R}^2 - \{0\}$ via the map $(x, y, z) \mapsto (x, y)$, and this projection defines a covering space $p: S \rightarrow \mathbb{R}^2 - \{0\}$ since for each open disk U in $\mathbb{R}^2 - \{0\}$, $p^{-1}(U)$ consists of countably many disjoint open disks in S , each mapped homeomorphically onto U by p .

Another example is the map $p: S^1 \rightarrow S^1$, $p(z) = z^n$ where we view z as a complex number with $|z| = 1$ and n is any positive integer. The closest one can come to realizing this covering space as a linear projection in 3-space analogous to the projection of the helix is to draw a circle wrapping around a cylinder n times and intersecting itself in $n - 1$ points that one has to imagine are not really intersections. For an alternative picture without this defect, embed S^1 in the boundary torus of a solid torus $S^1 \times D^2$ so that it winds n times monotonically around the S^1 factor without self-intersections, then restrict the projection $S^1 \times D^2 \rightarrow S^1 \times \{0\}$ to this embedded circle. The figure for Example 1.29 in the preceding section illustrates the case $n = 3$.

